UAV-GAP | CORE RESEARCH FINDINGS

Dual variability models / measured resolution contrasts / size-dependent effects

FM_P | DEGRADATION

46,368

25 leaves / 5 constraints

FM_D | DETECTOR

635,904

34 leaves / 4 constraints

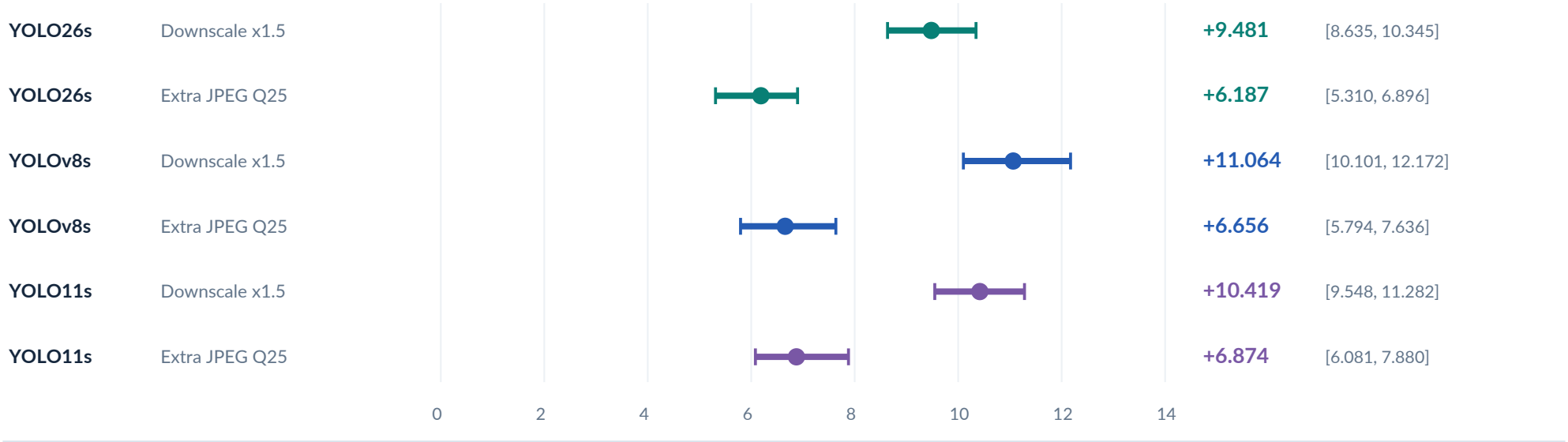
THEORETICAL CONFIGURATION PAIRS

29.49B

Logical combinations; not executed

NEW MEASURED DEVELOPMENT RESULTS | SAME-CHECKPOINT RESOLUTION GAIN

AP50:95 at 1280 minus 640 (percentage points; image-paired 95% intervals)



FINDING **Higher resolution improves overall AP in all six measured contrasts.**

Small-object gains exceed tiny-object gains; large-object AP decreases in all six settings.

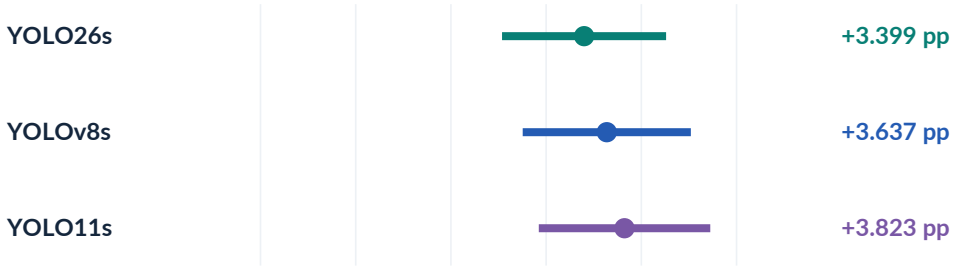
Source: CV261207 / VM26 verified reports; developmental findings, not independent test evidence.

UAV-GAP | CORE RESEARCH FINDINGS

Historical interaction / performance thresholds / evidence limits

DEVELOPMENT INTERACTION | RESOLUTION x BLUR x JPEG

Historical three-way contrast in AP50:95 (pp)



All three simultaneous 95% intervals exclude zero (300-image development cohort).

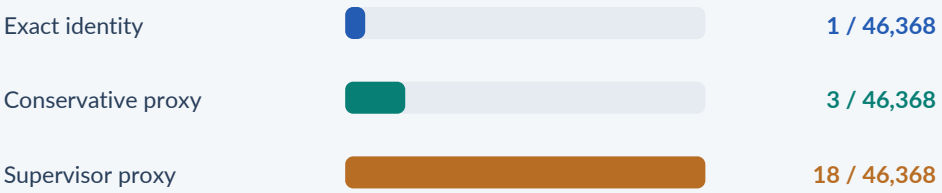
MEASURED SUBSET | TINY AP + PIPELINE LIMIT

Tiny AP >= 0.10, mean cached pipeline <= 100 ms

	1280 | DS	1280 | J25	640 | DS	640 | J25
YOLO26s	PASS	PASS	FAIL	FAIL
YOLOv8s	PASS	UNCERTAIN	FAIL	FAIL
YOLO11s	PASS	UNCERTAIN	FAIL	FAIL

4 PASS / 6 FAIL / 2 UNCERTAIN | measured subset only

STATIC COVERAGE IS POLICY-DEPENDENT



Training-augmentation coverage, not a measured detection failure rate.

CURRENT EVIDENCE BOUNDARY

RQ1 Two UVL models: validated logic; limited execution coverage.

RQ2 Measured development effects; no independent generalization.

RQ3 Subset PASS/FAIL; no full-space empirical-gap estimate.

RQ4 Clean-control canary passed; augmentation benefit untested.

248 image IDs / 247 unique byte hashes; formal test-dev: 0 / 500.

Source: CV261207 / VM26 verified reports; developmental findings, not independent test evidence.